M5470 : Introduction to Parallel Scientific Computing

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Homework 3

Due Date and Time: 4 March 2025, 11 pm

Question 1: Study the serial code for solving the heat conduction problem using second-order central-differencing scheme and the implicit Euler time-stepping scheme. In particular, pay attention to the Gauss-Seidel method for solving the system of linear equations. Run the code as given for $n = 800$ and $t_{en} = 0.001$. The solution at the final time step is written to the file 'T_xy_1277.dat'. You can visualize the solution by running 'plot_contours.m' in Matlab. This generates the files 'cont_T_1277.png' and 'line_midy_T_1277.png'.

(a) Write the serial version of the Alternating Direction Implicit (ADI) method to solve the system of linear equations. Ensure that it works correctly by comparing the solution at non-dimensional time $t \approx 0.001$ to the solution obtained using the original code provided to you. You can plot contours of T and line plots along the $y-$ centerline as above.

(b) Parallelize the ADI solver by introducing appropriate OpenMP directives. Briefly explain the parallelization strategy you selected. Ensure you get the same results for $p = 2, 4$ processors by plotting contours and lines as previously.

(c) Write the serial and parallel versions of the Red-Black Gauss-Seidel method. Once again, verify that the serial and parallel versions give identical answers to each other and similar answers to the original Gauss-Seidel solver. You should use a combination of contour and line plots to demonstrate correctness of your implementation.

In all, you should show 4 figures. Figure 1 should have the contour plots at $t \approx 0.001$ obtained from the original serial code, the ADI serial code, ADI parallel code with $p = 2$ and $p = 4$. Figure 2 should be a single plot with different lines corresponding to the same four versions as in Figure 1. Figures 3 and 4 should be similar to Figures 1 and 2 except that ADI solver is to be replaced by the Red-Black Gauss-Seidel solver.

Question 2: Launch p processors. Read in only a portion of the matrix stored in 'mat_xxxxxx.in' where 'xxxxxx' stands for the size of the matrix, n. Note that each processor will read in only n^2/p elements. Each processor reads in a vector from the file 'vec_xxxxxx_vecnum.in' entirely (i.e. all n elements are stored on each processor). The code checks if the vector is an eigenvector of the matrix and prints 'No' or 'Yes' followed by the eigenvalue if applicable.

If written correctly, the only MPI function needed to achieve this is MPI_Reduce. Note that the matrix can be distributed among the processors along rows, columns or both. Think carefully which is the best way that reduces the need for communication (i.e. the number of times MPI_Reduce needs to be called).

General Instructions:

1. Please prepare a short report with any plots or figures you generated with very brief (2 or 3 lines) comments. Do not waste time and effort in re-typing the question.

2. Submit the pdf on google classroom.

3. Upload your codes as well as the report pdf to Github.